

DEPARTMENT OF PHYSICS
MAR ATHANASIUS COLLEGE (AUTONOMOUS)
KOTHAMANGALAM

NAAC accredited 'A+' Grade Institution

92nd position in NIRF ranking 2025



PROJECT REPORT ON

**ELECTROCHEMICAL SYNTHESIS OF TRANSITION
METAL DICHALCOGENIDE-BASED MATERIALS FOR
SUPERCAPACITOR APPLICATIONS**

Submitted in partial fulfilment of the requirements for the degree of

MASTER OF SCIENCE IN PHYSICS

by

HARITHA P H

Under the supervision of

Prof. M. M. SHAIJUMON

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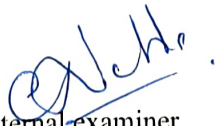
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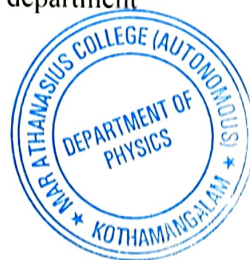
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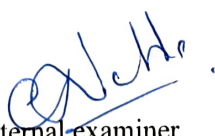
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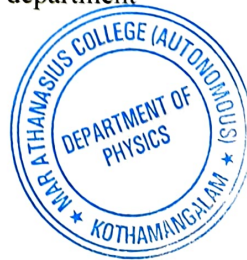
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MAR ATHANASIUS COLLEGE (AUTONOMOUS)

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DEPARTMENT OF PHYSICS

2025 - 2026



CERTIFICATE

This is to certify that the project work entitled “**ELECTROCHEMICAL SYNTHESIS OF TRANSITION METAL DICHALCOGENIDE-BASED MATERIALS FOR SUPERCAPACITOR APPLICATIONS,**” is a bona-fide record of work carried out by **HARITHA P H** (Reg. No: 24PPH1024) under the guidance and supervision of Prof. M. M. SHAIJUMON, Professor and Head, School of Physics, Indian Institute of Science Education and Research Thiruvananthapuram, in partial fulfilment of the requirements for the MSc in Physics at Mar Athanasius College (Autonomous), affiliated to Mahatma Gandhi University, during the academic year 2025 – 2026.

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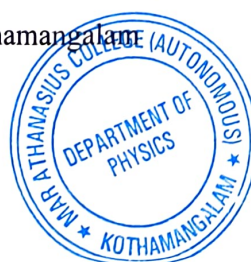
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DECLARATION

I hereby declare that the project report entitled “**ELECTROCHEMICAL SYNTHESIS OF TRANSITION METAL DICHALCOGENIDE-BASED MATERIALS FOR SUPERCAPACITOR APPLICATIONS,**” submitted in partial fulfilments of the requirements for the award of the degree of Master of Science in Physics of Mahatma Gandhi University, Kottayam, during the academic year 2025 – 2026, is a bona-fide work carried out by me under the supervision of Prof. M. M. SHAIJUMON, Professor and Head, School of Physics, Indian Institute of Science Education and Research Thiruvananthapuram.



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Place: Kothamangalam

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ABSTRACT

The increasing global energy demand necessitates the importance of an advanced energy storage system capable of ensuring the efficient and reliable supply of clean energy. This, in turn, demands the development of advanced materials with superior electrochemical properties for next-generation energy storage technologies. Herein, a simple and scalable synthesis route for transition metal dichalcogenide-based materials for supercapacitor applications is explored. Specifically, mixed-phase 1T/2H MoS₂ nanosheets were rapidly synthesized via microwave-assisted chemical exfoliation. Binder-free supercapacitor electrodes were fabricated by electrophoretic deposition of the exfoliated nanosheets onto carbon paper substrates, and their electrochemical performance was evaluated using cyclic voltammetry, galvanostatic charge-discharge, and electrochemical impedance spectroscopy measurements. The intrinsic capacitive behavior of the synthesized material was investigated in a three-electrode configuration, while the practical feasibility of the material was demonstrated through the fabrication of a symmetric coin-cell supercapacitor device. The electrophoretically deposited MoS₂ nanosheet electrodes exhibited a high specific capacitance of 324 F g⁻¹ at a current density of 2 A g⁻¹ in 1 M Na₂SO₄ aqueous electrolyte. Furthermore, the assembled symmetric coin-cell device delivered an energy density of 13.9 Wh kg⁻¹ at a current density of 1 A g⁻¹ and a power density of 0.4 kW kg⁻¹, while retaining 93% of its initial capacitance after 10,000 charge-discharge cycles at 5 A g⁻¹. The enhanced supercapacitor performance is attributed to the coexistence of metallic 1T and semiconducting 2H phases, improved electrical conductivity, and the binder-free electrode architecture achieved through electrophoretic deposition.

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CHAPTER 1

INTRODUCTION

1.1 General Introduction

The significant increase in both population and economic expansion has increased the global energy demand. Relying only on fossil fuels to meet the demand would be a major environmental concern due to the increase in greenhouse gases, climate change, air pollution, and ecological challenges.¹ Thus, a shift from the conventional energy sources to the renewable energies, such as solar energy, wind energy, tidal energy, and hydrogen fuels, is necessary.^{2,3} However, there are considerable downsides to these renewable energies, such as intermittency, unpredictability, costliness, geographical limitations, and large land requirements. So, there comes the need for advanced storage systems. There are different types of energy storage systems, including electrochemical energy storage systems such as batteries, supercapacitors, and conventional capacitors. Mechanical energy storage systems include pumped-hydro storage, flywheels, and compressed air. A thermal energy storage system stores excess heat or cold to be used later for industrial purposes. Chemical energy storage systems convert chemical energy into fuels like green hydrogen through electrolysis.⁴

Among the energy storage systems, the electrochemical energy storage systems hold a dominant advantage over other methods due to their high round-trip efficiency, high energy density, geographical independence, and instantaneous response speed.⁵ However, energy storage systems continue to face significant challenges in balancing key performance metrics. An ideal energy storage device should simultaneously possess high energy density, high power density, long cycle life, and low cost.⁶ No single energy storage technology satisfies all requirements. Therefore, the development of advanced electrode materials with improved electrochemical properties is essential for next-generation energy storage applications.

Focusing on the advanced energy storage solutions, the work on chemical and electrochemical materials is increasing tremendously, targeting several applications, including supercapacitors, batteries, portable electronics, and sensors, through various synthesis techniques to enhance the performance in energy storage.⁷ Recently, two-dimensional (2D) materials have emerged as promising candidates for energy storage

applications due to their exceptional physicochemical properties. Materials such as graphene, transition metal dichalcogenides (TMDs), MXenes, metal oxides, black phosphorus (BP), hexagonal boron nitride (h-BN), layered double hydroxides (LDHs), covalent organic frameworks (COFs), and metal-organic frameworks (MOFs) offer advantages including high surface area, fast electron transport, short ion diffusion pathways, and abundant active sites.⁸ (Figure 1.1) These unique characteristics make 2D materials highly attractive for advanced electrochemical energy storage systems.⁸

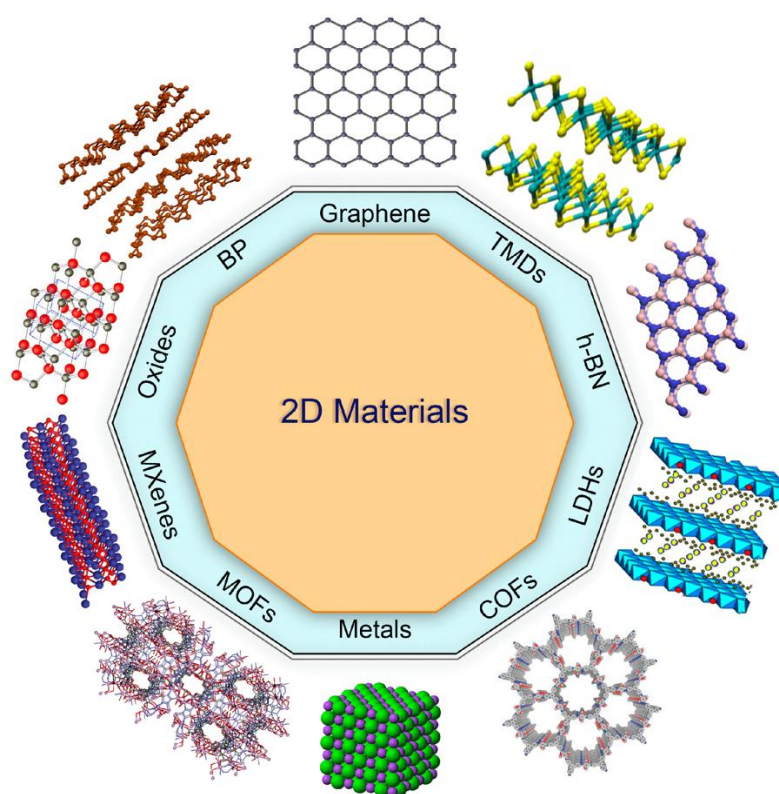


Figure 1.1 Examples of 2D materials. Reproduced from Ref. ⁸

1.2 Transition Metal Dichalcogenides (TMDs)

TMDs are prominent among 2D materials due to their unique mechanical, structural, and electrical properties, including stability, compositional tunability, porosity, electronic conductivity, high surface area, and layered architecture. Among them, MoS₂ has attracted significant attention due to its abundance, low cost, and promising electrochemical performance.⁹ TMDs possess the general chemical formula of MX₂ (X-M-X), where a layer of transition metal atoms (M) is sandwiched between two layers of chalcogen atoms (X).⁸ A single TMD layer consists of strongly bonded atoms connected through intralayer

covalent bonding, and adjacent layers in bulk TMD layers are held together by weak interlayer Van der Waal forces, making these materials versatile for advanced applications such as energy storage, optoelectronics, and flexible electronics.

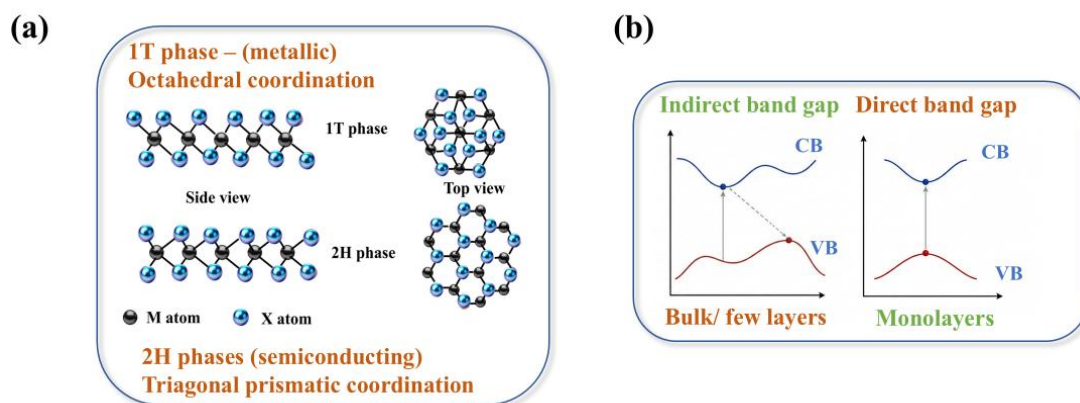


Figure 1.2 (a) Schematic of the crystal structure of 1T and 2H phases of MoS₂ (b) Direct and indirect bandgap of bulk and monolayer MoS₂. Reproduced from Ref. ¹⁰

When bulk or few-layered TMDs such as MoS₂ are thinned down to a monolayer, their electronic structure often changes from an indirect to a direct bandgap, leading to enhanced electronic and optical properties (Figure 1.2b). Among the various polymorphs of TMDs, the most widely studied phases are the metallic 1T phase (octahedral coordination) and the semiconducting 2H phase (triagonal prismatic coordination).⁸ The 1T phase exhibits high electrical conductivity owing to its octahedral coordination geometry (Figure 1.2a); however, it is thermodynamically metastable. In contrast, the 2H phase is the thermodynamically stable semiconducting phase with a trigonal prismatic coordination structure.⁸ Different phases of TMDs can be selectively obtained through various synthesis methods. Their layered structure facilitates efficient ion transport, while their high surface area, multiple accessible oxidation states that contribute to pseudocapacitance, and tunable electronic properties make them highly attractive for electrochemical energy storage applications.

1.3 Synthesis Methods Of TMDS

Various methods have been developed for the synthesis of TMD materials, which are broadly classified into top-down and bottom-up approaches. Each method offers distinct advantages and limitations, but precise control over the synthesis route can directly impact

the resulting properties and thereby determine their suitability for specific applications. The conventional top-down techniques include liquid phase exfoliation, mechanical exfoliation, and sputtering. In contrast, bottom-up approaches include chemical vapor deposition, hydrothermal synthesis, sol-gel process, and atomic layer deposition.¹¹ However, many of these sophisticated synthesis techniques require extreme processing conditions, such as high temperatures, vacuum, and inert atmospheres, to achieve controlled material growth. While these methods excel at producing high-quality nanomaterials under laboratory conditions, their scaling up will be challenging and expensive. Therefore, microwave-assisted chemical exfoliation combined with electrophoretic and electrochemical deposition was selected as the synthesis strategy in this project. The integration of rapid exfoliation and low-cost deposition approaches enables a simple and scalable route for the synthesis of TMD materials for energy applications.

1.4 Applications

In the focused vision to meet the global energy demand, it is essential to pursue the sustainable, renewable, and sources of clean energy.¹² Owing to their unique structural, electronic, and electrochemical properties, TMDs have attracted significant attention across a wide range of applications.^{13,14} The major applications in the energy harvesting and storage sector includes, supercapacitors, batteries, and fuel cells. Beyond energy applications, TMDs have also shown great potential in nanoelectronics, optoelectronics, biomedicine, sensing, and environmental remediation, making them highly versatile materials in modern nanotechnology.

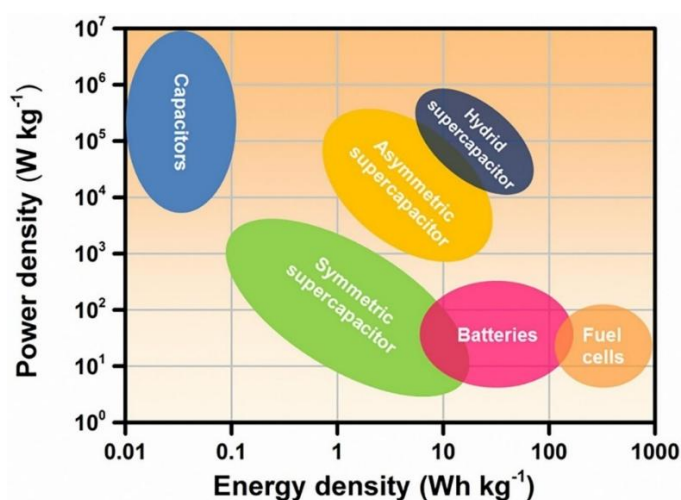


Figure 1.3 Ragone plot illustrating the relationship between specific power density and energy density for various electrical energy storage technologies, reproduced from Ref.¹⁵

One of the widely researched applications in the energy storage sectors, supercapacitors show great progress due to their high-power density, long cycle life, and ultrafast charge-discharge rate.^{16–18} Supercapacitors are attractive energy storage devices owing to their outstanding performance, simple operating mechanism, and ease of integration into electronic systems. As shown in the Ragone plot (Figure 1.3), supercapacitors position itself in between the traditional capacitors and rechargeable batteries by combining high power density with moderate energy density (Figure 1.4 a, b). Future advancements in energy storage technologies are focused on bridging the gap between batteries and conventional capacitors by simultaneously achieving high energy density and high-power density.¹⁷ Consequently, the region above current supercapacitor performance in the Ragone plot is considered the target area for next-generation energy storage devices. It charges and discharges rapidly, giving high bursts of power and has a long-life cycle without degrading, but stores less energy than a battery.⁶

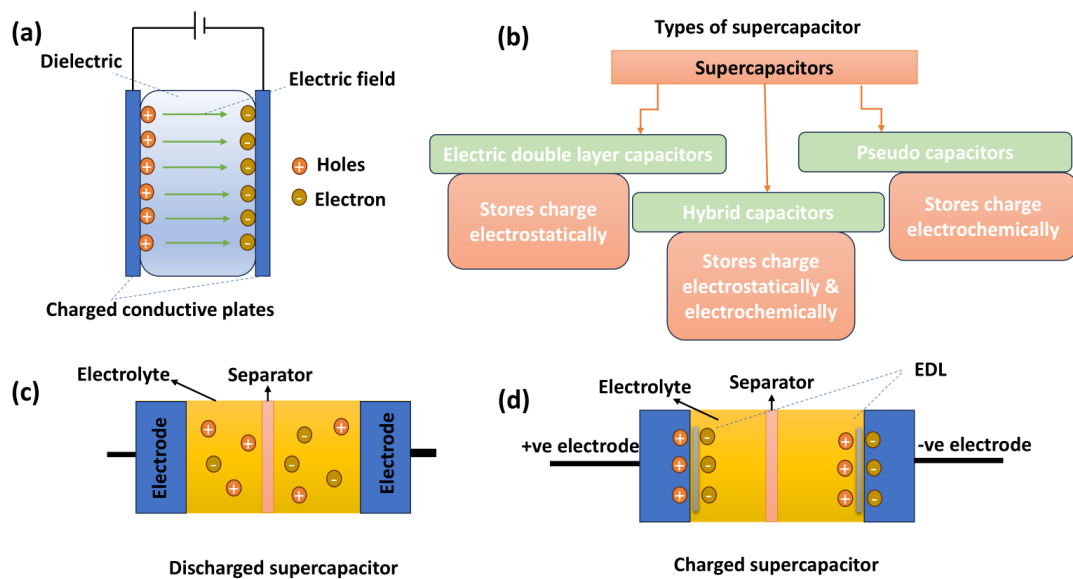


Figure 1.4 (a) Schematic of a conventional capacitor when charging, (b) types of supercapacitors. Schematic of a supercapacitor at (c) discharged and (d) charged mode.

The energy storage capability of supercapacitors arises mainly from two mechanisms: Electrostatic double-layer capacitance (EDLC) and Pseudocapacitance.⁶

a) Electric double-layer capacitance – electrostatically stores the charges at the interface of the capacitor electrode. In a supercapacitor, there are two electrodes dispersed in an electrolyte and connected to a power supply. When a constant voltage is applied to a

supercapacitor, the ions dissolved in the liquid electrolyte move towards the oppositely charged electrode (Figure 1.4 c, d). Positively charged ions (cations) gather near the negative electrode, and negatively charged ions (anions) gather near the positive electrode; this forms the electric double layer, storing energy physically rather than chemically.⁶

b) Pseudocapacitance – faradaically storing charges at the electrode surface via surface reactions. Pseudo capacitance is governed by electron charge transfer between the electrode and electrolyte by means of redox reactions (reduction & oxidation). In pseudo-capacitors, the oxidation-reduction reaction occurs between the electrode and the electrolyte solution and hence results in a special capacitance.⁶

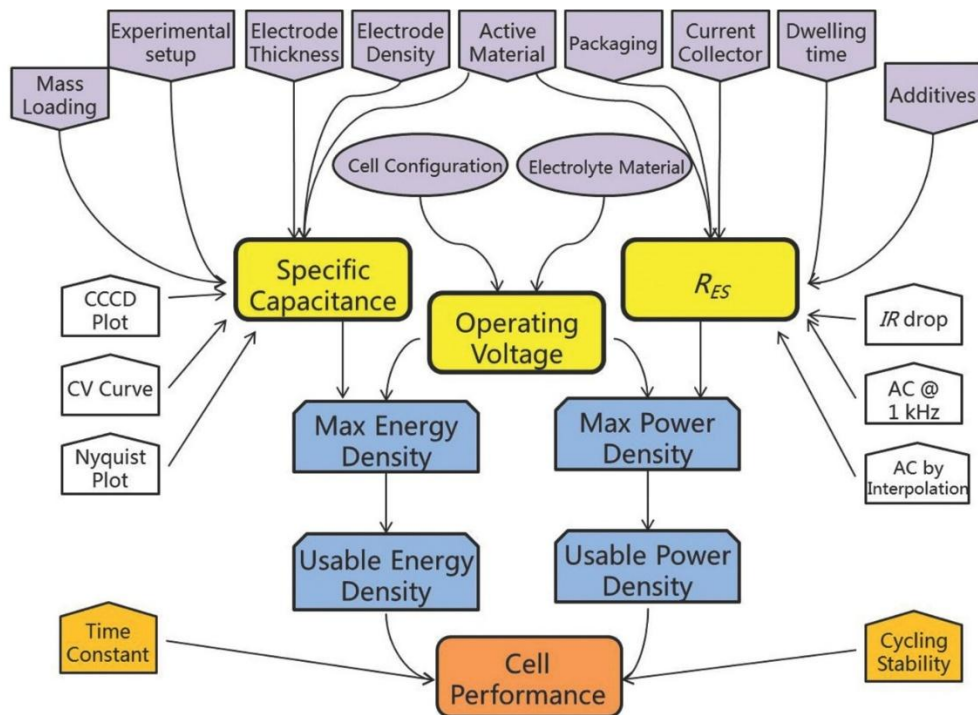


Figure 1.5 Illustration of the key performance parameters, characterization techniques, and major factors affecting the evaluation of supercapacitors, reproduced from Ref..⁶

The performance of supercapacitors is analyzed by electrochemical characterization methods and key performance metrics, including cyclic voltammetry (CV), galvanostatic charge–discharge (GCD), electrochemical impedance spectroscopy (EIS), and cycling stability tests, as well as being strongly influenced by the choice of electrode materials and synthesis routes (Figure 1.5). Achieving high capacitance, energy density, power density,

long cycle life, and low cost require the development of suitable electrode materials and optimized fabrication strategies.⁶

1.5 Objectives and Scope of the Work

This work aims to explore the simple and scalable synthesis routes of TMD-based materials for energy storage applications. The specific objectives are:

- To synthesize mixed-phase 1T/2H MoS₂ nanosheets through microwave-assisted chemical exfoliation.
- To fabricate binder-free supercapacitor electrodes by electrophoretic deposition of MoS₂ nanosheets onto carbon paper.
- The electrochemical performance of the fabricated electrodes is systematically evaluated in both three-electrode and symmetric coin-cell configurations using aqueous electrolytes.

The simplicity, scalability, and high yield of the selected synthesis and deposition methods provide significant opportunities for extending this work toward the fabrication of TMD-based materials for high-performance micro-supercapacitor applications. Future studies can focus on optimizing the deposition of these materials and their heterostructures onto interdigitated electrode architectures to enhance the performance of symmetric micro-supercapacitors.

CHAPTER 2

MATERIALS AND METHODS

MoS₂ nanosheets were synthesized via a microwave-assisted chemical exfoliation route, followed by electrophoretic deposition onto conductive substrates to fabricate supercapacitor electrodes. In parallel, the electrodeposition of various TMD materials is also being explored to develop nanostructured electrodes with enhanced electrochemical performance.

2.1 Materials

Molybdenum sulfide powder (MoS₂, Sigma-Aldrich), Acetylene black, *n*-Butyllithium (1.6M in hexane), Hexane (Rankem), Sodium sulfate ($\geq 99.0\%$, anhydrous powder, Sigma-Aldrich), Ammonium tetrathiomolybdate ((NH₄)₂MoS₄, 99.95%, Alfa Aesar), potassium chloride (99.5%, Finar reagents), thiourea (99%, Alfa Aesar) and cobalt (II) chloride hexahydrate (98%, Sigma-Aldrich).

2.2 Microwave-Assisted Chemical Exfoliation (MWCE) of MoS₂

MoS₂ nanosheets were synthesized via a microwave-assisted chemical exfoliation method, following the literature.¹⁹ Conventional chemical exfoliation is typically a low-yield, multistep, and time-consuming process, often requiring more than a day to produce milligram quantities of material.¹⁹ In contrast, MWCE offers a rapid and scalable alternative, enabling the synthesis of TMD nanostructures within a shorter time while simultaneously producing mixed metallic and semiconducting phases that are beneficial for electrochemical applications.

Briefly, 300 mg of MoS₂ powder was thoroughly mixed with 100 mg of acetylene black powder using a mortar and pestle for 15 min. The mixture was then transferred to a microwave glass vial, followed by adding 15 mL of hexane and 3 mL of *n*-butyllithium solution. The reaction vessel was irradiated with 2.45 GHz microwave radiation for 30s at a constant maximum power of 800 W. During irradiation, the mixture was magnetically stirred at 240 rpm, while the reaction temperature was maintained at 150°C. This treatment facilitated the intercalation into the MoS₂ layers, and the resulting Li-intercalated MoS₂ (Li_xMoS₂) product was collected after cooling to room temperature. The product was

washed twice with hexane to completely remove residual n-butyllithium by centrifugation at 4000 rpm for 15 min. The washed product was then dried in a fume hood for 10 min. Subsequently, the Li_xMoS_2 was exfoliated in deionized (DI) water by ultrasonication for 30 min. The resulting suspension was centrifuged at 4000 rpm for 15 min to remove acetylene black and any unreacted bulk MoS_2 . The supernatant containing dispersed microwave-assisted chemically exfoliated MoS_2 (MWCE MoS_2) nanosheets was carefully collected after centrifugation at 3000 rpm for 15 min. The resulting thick black dispersion of MoS_2 nanosheets was used directly for electrophoretic deposition.

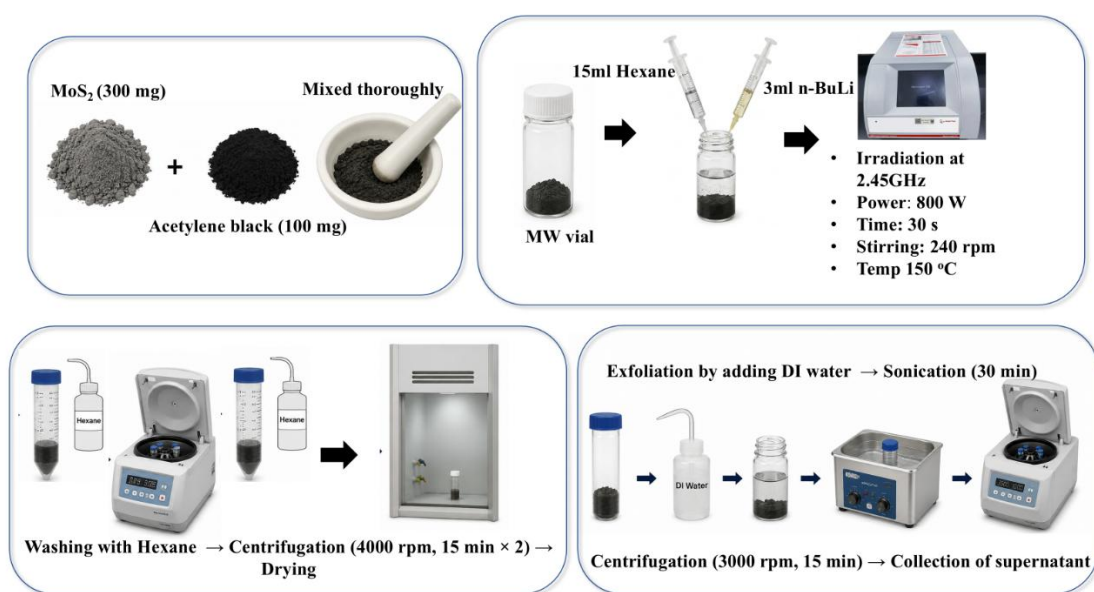


Figure 2.1 Schematic illustration of the synthesis of MWCE of MoS_2

2.3 Electrophoretic Deposition of MWCE MoS_2

Electrophoretic deposition is a physical process that relies on charge particles present in the suspensions. Electrophoretic deposition is a versatile colloidal processing technique used to produce thin films and coatings. The mechanism in EPD facilitates the charged particles (or macromolecules) to move in a suspension towards the oppositely biased electrode, followed by the orderly deposition on the conducting substrate.^{20,21} The ability to produce uniform coatings on complex substrates, facilitate patterned electrode fabrication, and enable simple, scalable, and low-cost manufacturing makes EPD a versatile and attractive fabrication technique for energy storage applications.²²

Supercapacitor electrodes were fabricated by EPD of MWCE MoS₂ nanosheets onto carbon paper (5 mm × 5 mm). A platinum plate was employed as the counter electrode, and a constant potential of 10 V was applied for 10 min. During the deposition process, the MoS₂ nanosheets (negatively charged) migrated under the applied electric field and were deposited onto the positively biased electrode. After deposition, the coated carbon paper was rinsed with DI water and isopropyl alcohol (IPA) to remove loosely bound particles and residual impurities. The electrodes were subsequently dried in an oven at 60 °C overnight.

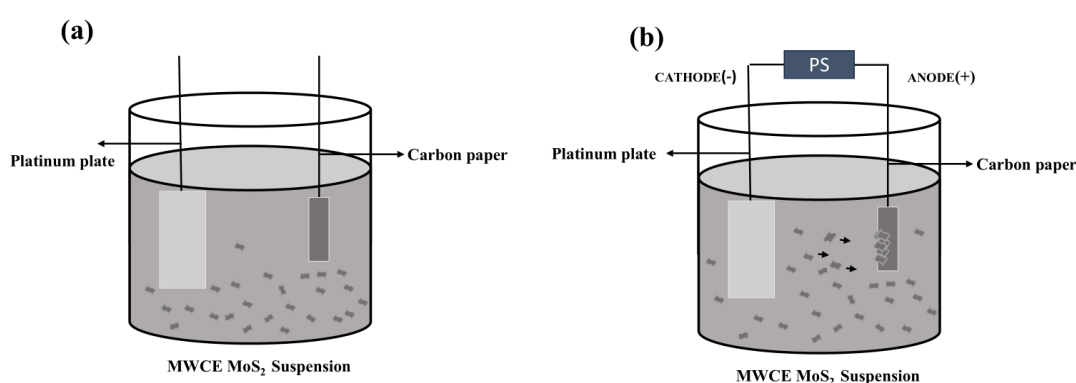


Figure 2.2 Schematic illustration of EPD, (a) unbiased and (b) biased setup.

2.4 Electrodeposition of TMDs

The mechanism of electrodeposition is based on an electrochemical synthesis process that enables the direct deposition of materials onto a desired conducting substrate through oxidation and reduction reactions occurring at the electrode–electrolyte interface upon the application of an external potential.

2.4.1 Electrodeposition of MoS₂

MoS₂ thin films were electrodeposited onto indium tin oxide (ITO)-coated glass substrates using an aqueous electrolyte containing 10 mM ammonium tetrathiomolybdate ((NH₄)₂MoS₄) and 0.2 M potassium chloride (KCl) dissolved in 10 mL of DI water. Before deposition, the ITO substrates were thoroughly cleaned by repeated rinsing with DI water and ethanol/IPA. Electrodeposition was carried out in a three-electrode configuration (Figure 2.3), with the ITO-coated glass serving as the working electrode, a platinum wire as the counter electrode, and an Ag/AgCl as the reference electrode.²² CV measurements

were first performed using the precursor solution to determine the reduction potential corresponding to MoS₂ formation. Based on the CV, MoS₂ deposition was subsequently carried out by chronoamperometry at an applied potential of −1.02 V for 10 min. A uniform thin film of MoS₂ was obtained on the ITO substrate, exhibiting good adhesion to the conductive surface. The as-deposited MoS₂ films were annealed at 300 °C for 2 h in a quartz tube furnace under an argon atmosphere to improve their crystallinity.

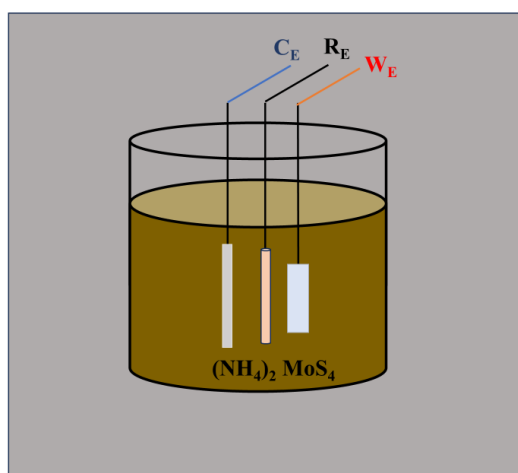
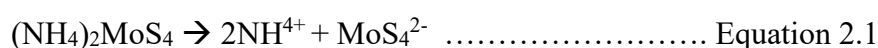


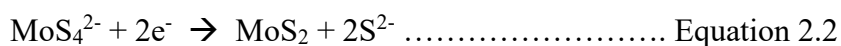
Figure 2.3 Schematic illustration of the electrodeposition process

The possible electrochemical redox reactions would be; ²²

(1) Dissociation of precursor



(2) Electrochemical reduction of tetrathiomolybdate



2.4.2 Electrodeposition of CoS

Cobalt monosulfide (CoS) thin films were electrodeposited using an aqueous precursor solution prepared by dissolving 100 mg of cobalt chloride and 200 mg of thiourea in 10 mL of DI water. Before deposition, CV measurements were carried out in a three-electrode configuration to determine the reduction potential corresponding to CoS formation. Based on the CV results, electrodeposition was subsequently performed by chronoamperometry

at an applied potential of approximately -1.1 V for 10 min. A thin layer of CoS was deposited on the conducting substrate, exhibiting good adhesion and uniform coverage. All synthesis procedures were performed multiple times to ensure reproducibility and to obtain a better understanding of the growth and deposition processes.

2.5 Materials Characterization

The morphology of the as synthesized materials was analysed using field emission scanning electron microscope (FE-SEM, Nova NanoSEM 450, FEI). The spectroscopical characterization was done by performing Raman spectroscopy using a 532nm laser using HORIBA XploraPlus Micro Raman spectrometer.

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Exfoliation of MoS₂ and Electrophoretic Deposition

3.1.1 Morphological and Spectroscopic Characterization

MoS₂ nanosheets were successfully exfoliated via microwave-assisted chemical exfoliation (MWCE), as shown in section 2.1. MWCE MoS₂ nanosheets were characterized by scanning electron microscopy (SEM) and Raman spectroscopy. The Raman spectra of bulk MoS₂ (Figure 3.1a) show the typical in-plane E_{2g}¹ mode at 380.7 cm⁻¹ and out-of-plane A_{1g} mode at 404.4 cm⁻¹ of the 2H phase of MoS₂. Figure 3.1b shows the Raman spectra of MWCE MoS₂ nanosheets, exhibiting the characteristic Raman modes of both the 1T and 2H phases. In the Raman spectrum of MWCE MoS₂, the peaks at 382.8 and 404.4 cm⁻¹ are assigned to the E_{2g}¹ and A_{1g} mode vibrational modes of the 2H-MoS₂ phase, respectively. In addition, the peaks at 156.5, 219, and 330.1 cm⁻¹ correspond to the J₁, J₂, and J₃ vibrational modes, confirming the presence of an additional 1T phase. The reduction in the peak separation between the E_{2g}¹ and A_{1g} mode of the 2H-MoS₂ phase from 23.7 to 21.5 cm⁻¹ indicates successful exfoliation of the MoS₂ layers.

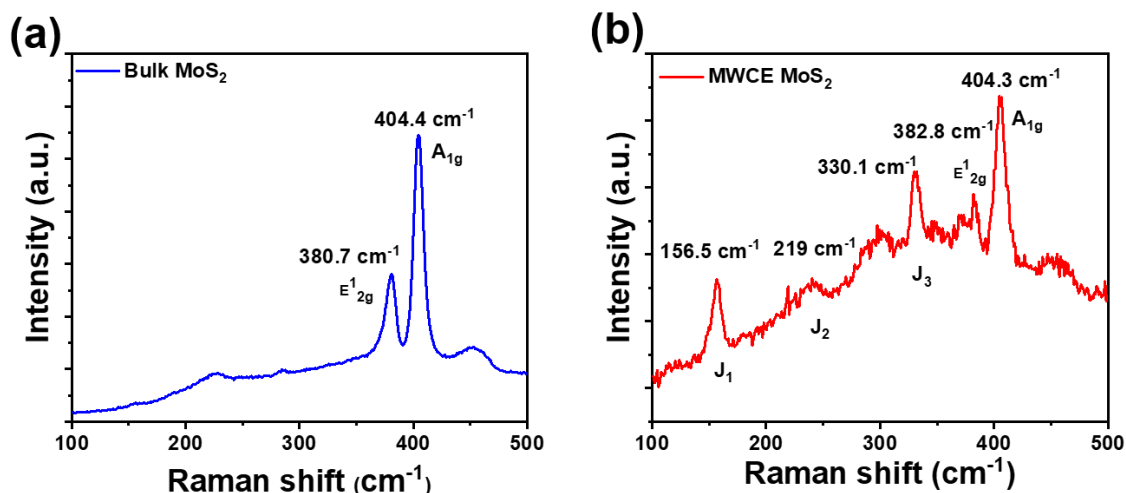


Figure 3.1 Raman spectra of (a) MoS₂ powder, and (b) MWCE MoS₂

SEM imaging was conducted to investigate the morphology of MoS₂ powder and MWCE MoS₂ samples coated in carbon paper. Figures 3.2a, b show the SEM images of MoS₂ powder coated on carbon paper, revealing bulk flakes. In contrast, Figures 3.2c, d display

the SEM images of MWCE MoS₂, which clearly reveal the presence of exfoliated nanosheets with a layered morphology.

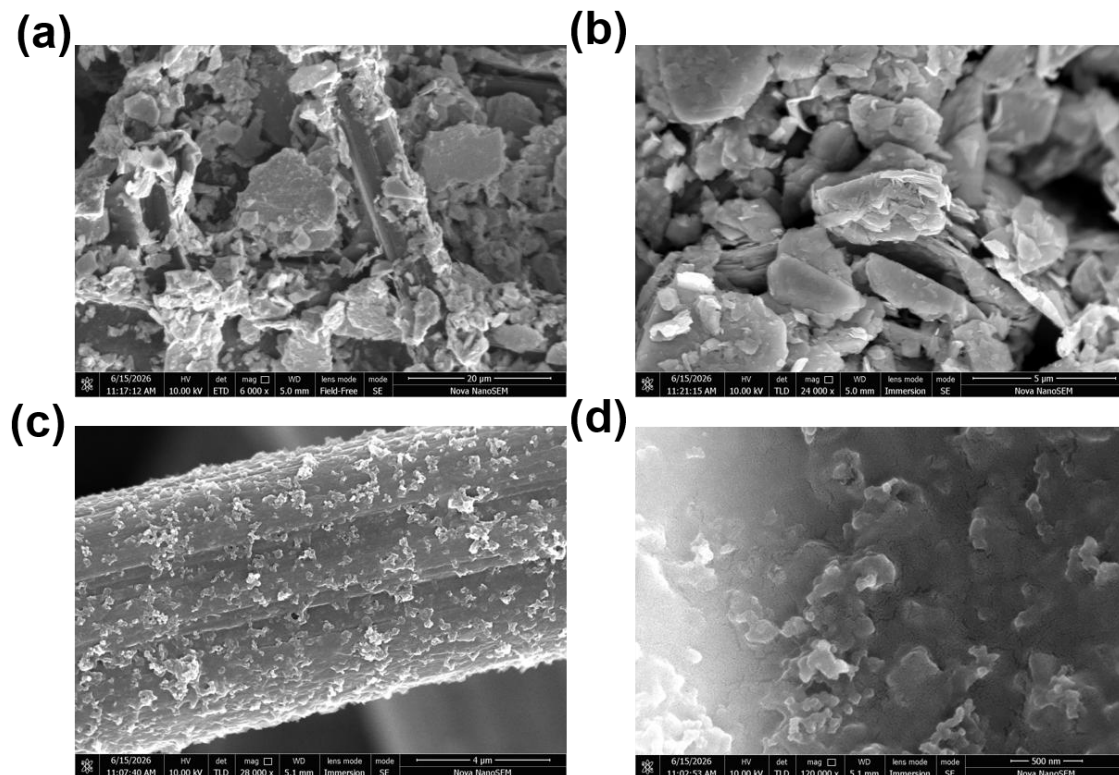


Figure 3.2 SEM images of (a, b) MoS₂ powder, and (c, d) MWCE MoS₂ nanosheets coated on carbon paper, recorded at different magnifications.

Figure 3.3a shows the Raman spectrum of electrophoretically deposited MWCE MoS₂ (EPD MWCE MoS₂) nanosheets, which exhibits characteristic Raman modes of both the 1T and 2H phases, similar to those observed for pristine MWCE MoS₂. The peaks located at 382.9 (E_{12g} mode) and 405.4 cm⁻¹ (A_{1g} mode) correspond to the 2H-MoS₂ phase, respectively. In addition, the peaks at 156.5, 217.9, and 329.5 cm⁻¹ are assigned to the J₁, J₂, and J₃ vibrational modes, respectively, confirming the presence of the metallic 1T phase. Figures 3b-d show the SEM images of electrophoretically deposited MWCE MoS₂ (EPD MWCE MoS₂) coated on carbon paper. The images reveal a nanosheet morphology similar to that observed for the pristine MWCE MoS₂ sample, indicating that the electrophoretic deposition process preserves the exfoliated layered structure of the nanosheets.

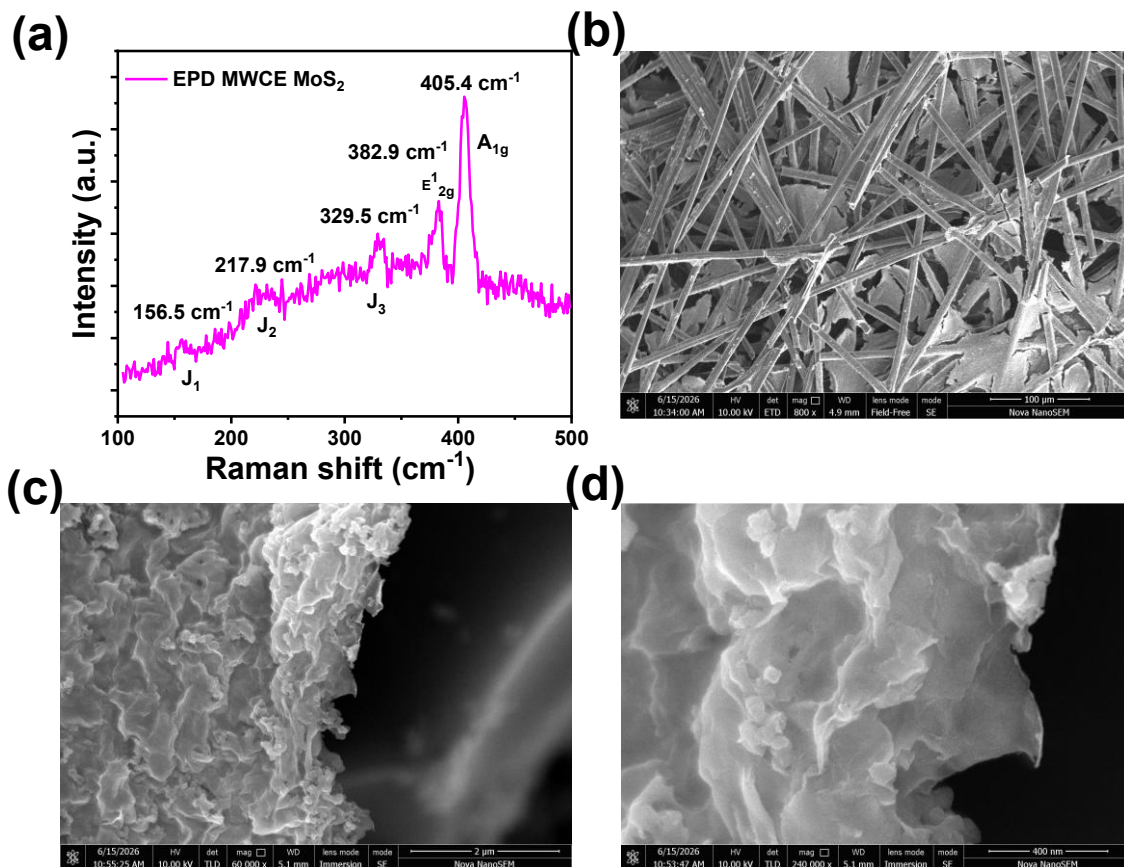


Figure 3.3 (a) Raman spectra and (b-d) SEM images of EPD MoS₂.

3.1.2 Electrochemical Measurements

3.1.2.1 Supercapacitance Measurements of EPD MoS₂

The electrochemical measurements of the samples were performed using a Biologic VMP3 electrochemical workstation in a standard three-electrode configuration, where the platinum wire served as the counter electrode, a silver/silver chloride (Ag/AgCl) electrode acted as the reference electrode, and EPD MoS₂ was taken as the working electrode. The electrochemical performance of EPD MoS₂ nanosheets on carbon paper electrodes (active area: 5 mm × 5 mm) was investigated using CV, GCD, and EIS measurements in a three-electrode configuration. All measurements were carried out in 1 M Na₂SO₄ aqueous electrolyte within a potential window of −0.3 to 0.6 V vs Ag/AgCl. The photograph of the three-electrode setup is shown in Figure 3.4a, while the comparative cyclic voltammograms of MoS₂ powder and EPD MoS₂ electrodes are presented in Figure 3.4b. The significantly larger CV area of the EPD MoS₂ electrode compared to the MoS₂ powder indicates its enhanced capacitive performance.

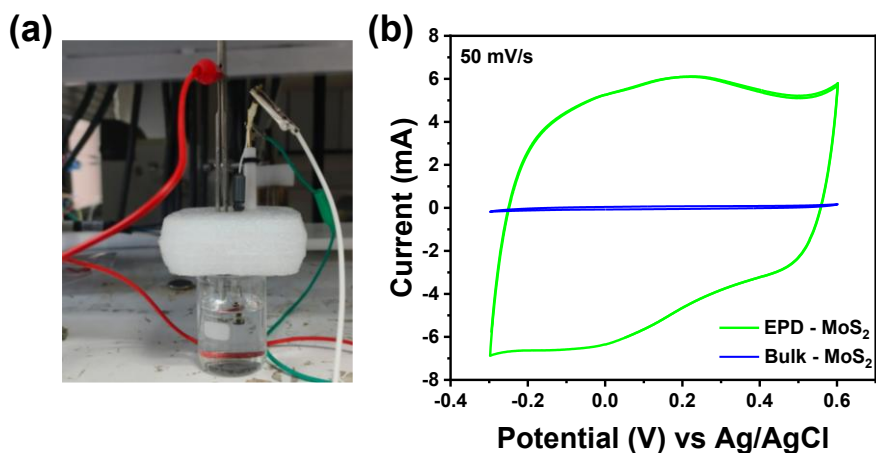


Figure 3.4 (a) Photograph of the three-electrode setup, (b) comparative CV profile for MoS₂ powder and the EPD MoS₂ electrodes

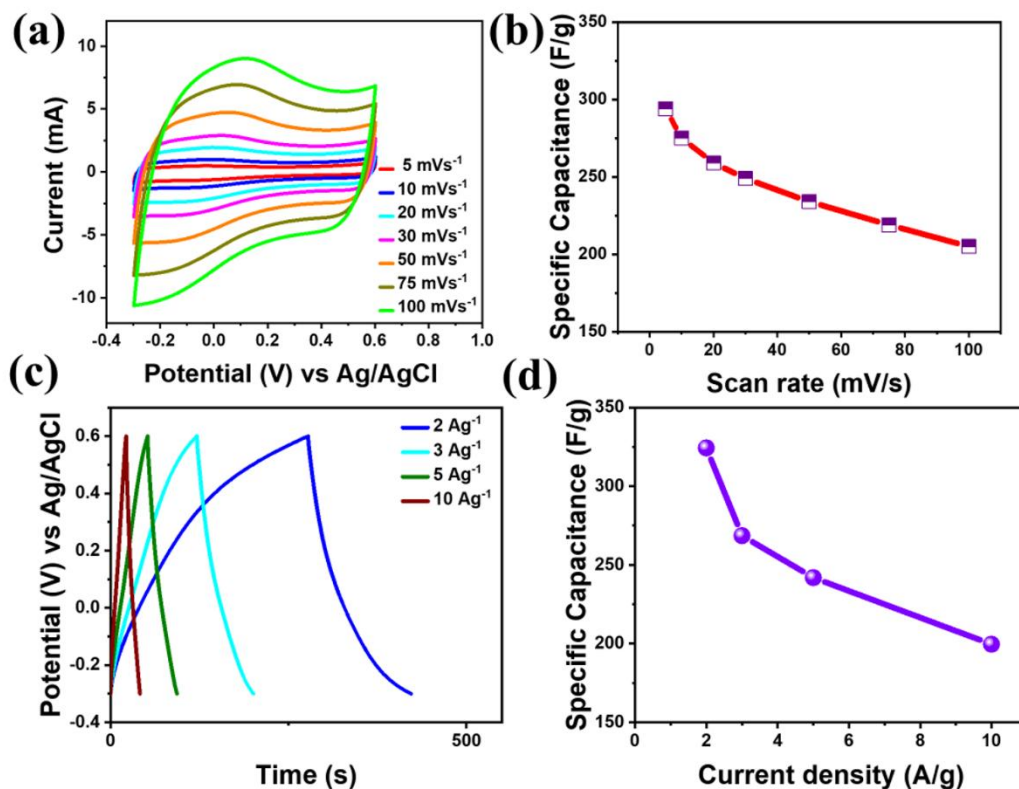


Figure 3.5 The electrochemical characterizations of EPD MoS₂ (a) CV curves at different scan rates, (b) corresponding plot of specific capacitance vs scan rates, (c) GCD cycles at different current densities, and (d) corresponding plot of specific capacitance vs current density.

Figure 3.5 presents the electrochemical characterization of the EPD MoS₂ electrode. Figure 3.5a shows the CV curves recorded at different scan rates, where the enclosed CV area

increases with increasing scan rate, indicating good capacitive behavior and charge storage capability. The specific capacitance (C_s) of EPD MoS₂ nanosheets was calculated from the CV curves according to the following equation;

$$C_s = \frac{A}{2v \times m \times \Delta V} \text{ Fg}^{-1} \quad \dots\dots\dots \text{Equation 3.1}$$

Where A denotes the integral area of the cyclic voltammogram loop, v is the scan rate, ΔV is the potential window and m is the mass of the electrode material (0.3 mg). Figure 3.5b shows the variation of specific capacitance with scan rate, with the electrode exhibiting a maximum specific capacitance of 294 F g⁻¹ at a scan rate of 5 mV s⁻¹, and a specific capacitance of 205.2 F g⁻¹ was obtained at a high scan rate of 100 mV s⁻¹.

The supercapacitor performance was further evaluated using GCD measurements at constant current densities. Figure 3.5c shows the GCD curves of the EPD MoS₂ electrode recorded at different current densities of 2, 3, 5, and 10 A g⁻¹. The specific capacitance measured from the GCD curve using the equation;

$$C_{\text{spec}} = \frac{I \Delta t}{m \Delta V} \text{ Fg}^{-1} \quad \dots\dots\dots \text{Equation 3.2}$$

Where I denote the constant current applied, ΔV is the potential window, Δt represents the discharged time and m is the mass of the electrode material. Fig 3.5d shows the graph between specific capacitance vs current density. The maximum current density of 324 F g⁻¹ was obtained at a current density of 2 A g⁻¹. The high specific capacitance can be attributed to the increased accessible surface area of the deposited MoS₂ nanosheets and their good electrical conductivity.

3.1.2.2 Symmetric Cell Capacitance Measurement of EPD MoS₂

To evaluate the practical applicability of the material, the capacitive performance of a symmetric supercapacitor based on EPD MoS₂ nanosheets was investigated in a two-electrode configuration, as shown in Figure 3.6a, with a mass loading of 0.4 mg (each electrode) in a 5 mm x 5 mm in 1 M Na₂SO₄ aqueous electrolyte. Figure 3.6b shows the CV curves at different scan rates. The absence of any redox peak in the CV curve for EPD MoS₂ nanosheets-based symmetric supercapacitor implies the dominance of the electric double-layer at the electrode-electrolyte interface.

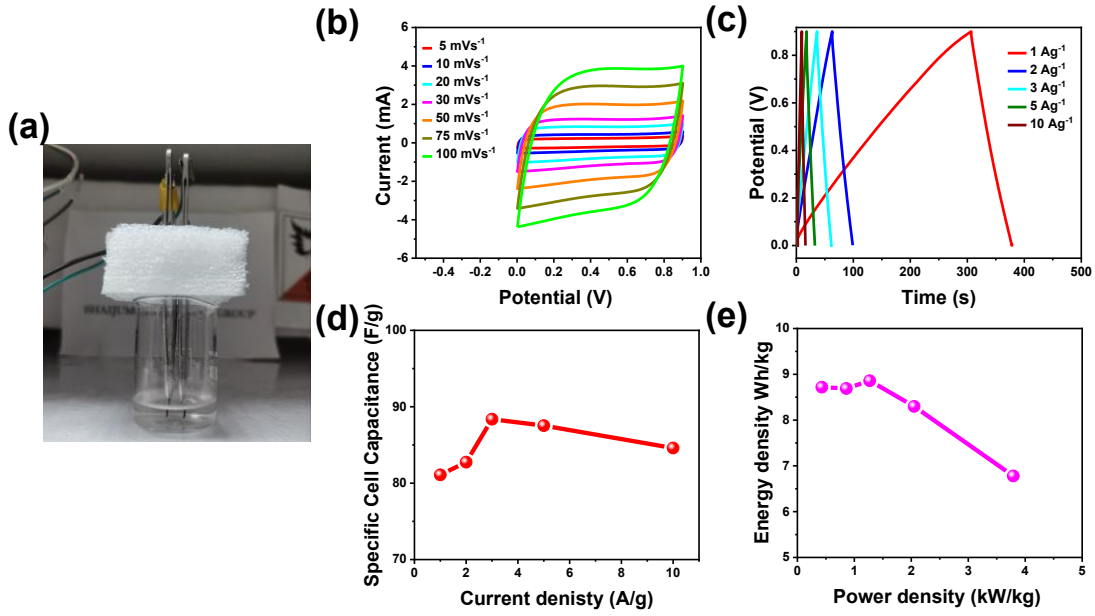


Figure 3.6 (a) Photograph of the two-electrode setup used for evaluating the cell capacitance of the symmetric supercapacitor device, (b) CV curves recorded at different scan rates, (c) GCD curves obtained at different current densities for the EPD MoS₂-based symmetric supercapacitor, (d) variation of cell capacitance with current density, and (e) Ragone plot showing the relationship between energy density and power density.

Figure 3.6c shows the GCD curves of the EPD MoS₂ electrode recorded at different current densities of 1, 2, 3, 5, and 10 A g⁻¹. The cell capacitance measured from the GCD curve using the equation;

$$C_{\text{cell}} = \frac{I\Delta t}{\Delta V} \text{ Fg}^{-1} \quad \dots\dots\dots \text{Equation 3.3}$$

$$C_s = \frac{I\Delta t}{m_{\text{total}}\Delta V} \text{ Fg}^{-1} \quad \dots\dots\dots \text{Equation 3.4}$$

Where I denote the constant current applied, ΔV is the potential window, Δt represents the discharged time and m represents the total mass of both the electrode materials. The maximum specific cell capacitance of 82.7 F g⁻¹ was obtained at the current density of 2Ag⁻¹. The lower cell capacitance observed for the symmetric device is attributed to the series

capacitance effect inherent to the two-electrode configuration and the additional mass contribution from the second electrode. However, the capacitance of a single EPD MoS₂ nanosheet electrode (half-cell) is estimated to be 330.8 F g⁻¹, which is approximately four times higher than the specific cell capacitance of the symmetric device. Further, energy density (E) and power density (P) of EPD MoS₂ nanosheets were calculated using the following equations;

$$E = C_S \frac{(\Delta V)^2}{7.2} \text{ Wh (kg)}^{-1} \dots\dots\dots \text{Equation 3.5}$$

$$P = \frac{E \times 3600}{\Delta t} \text{ kW (kg)}^{-1} \dots\dots\dots \text{Equation 3.6}$$

An energy density of 8.86 Wh kg⁻¹ at a power density of 0.86 kW kg⁻¹ was achieved for the symmetric full-cell supercapacitor at a discharge current density of 2 A g⁻¹, as shown in Figure 3.6e.

3.1.2.3 Coin Cell Fabrication and Stability Studies of EPD MoS₂

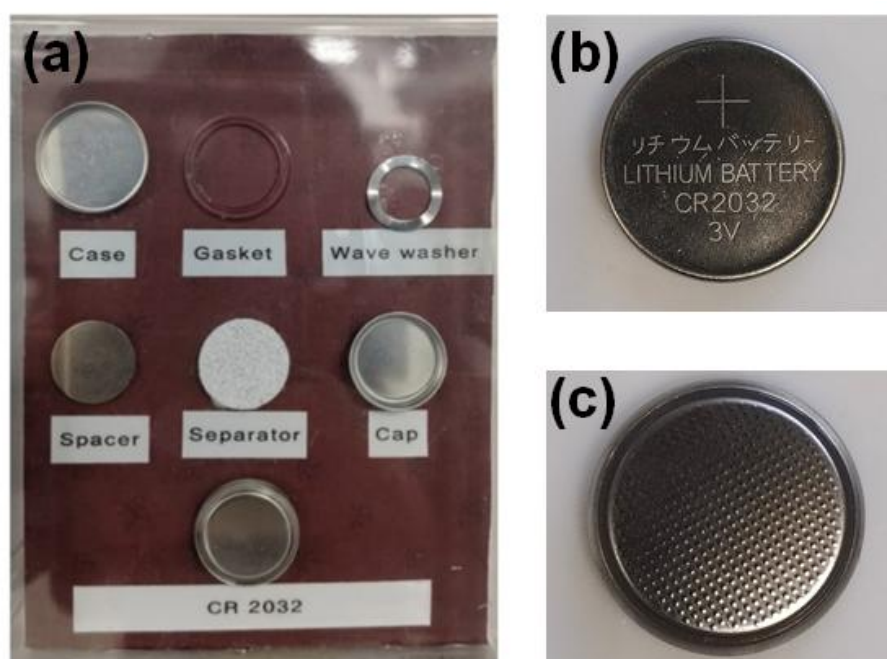


Figure 3.7 (a) The photograph of different parts of a coin cell, (b, c) the photograph of an assembled coin cell.

To investigate the real device feasibility, we studied the capacitive performance of EPD MoS₂ nanosheets by constructing a coin cell device. Figure 3.7a presents the different components used for coin-cell fabrication, whereas Figure 3.7b shows the fully assembled coin-cell supercapacitor.

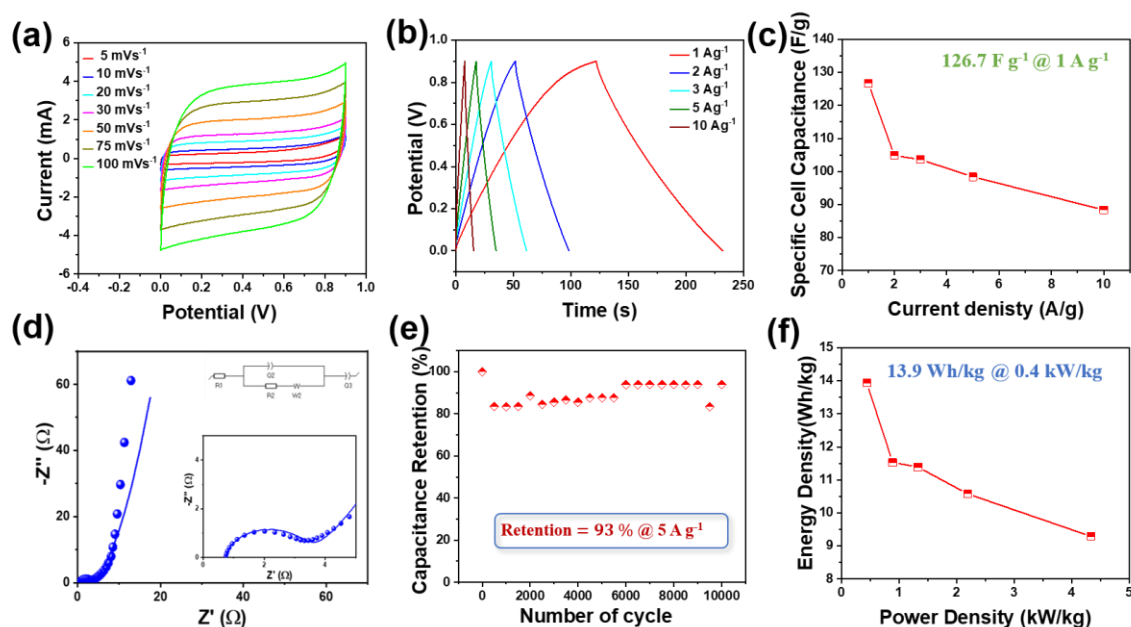


Figure 3.8 The electrochemical characterization of EPD MoS₂-based coin cell (a) CV curves at different scan rates, (b) GCD at different current densities, (c) specific cell capacity vs current density, (d) Nyquist plot, (e) cyclic stability test at 5 A g⁻¹ (f) Energy density vs Power density.

The electrochemical performance of the fabricated coin-cell supercapacitor was evaluated using CV, GCD, EIS, and cycling stability measurements. Figure 3.8a shows the CV curves of the coin-cell device recorded at different scan rates, demonstrating its capacitive behaviour over a range of operating conditions. The specific capacitance was calculated from the GCD measurements, as they provide a more reliable estimation than CV analysis. Figure 3.8b presents the GCD curves obtained at different current densities. A specific cell capacitance of 104.8 F g⁻¹ was achieved for the coin supercapacitor at a current density of 2 A g⁻¹, as shown in Figure 3.8c. Similar to the symmetric cell configuration, the lower capacitance of the coin cell arises from the series capacitance effect and the contribution of the total active mass from both electrodes. The equations used for calculating the specific capacitance from GCD measurements, as well as the energy density and power density, are identical to those employed for the symmetric supercapacitor device.

The Nyquist impedance plots, recorded over a frequency range from 100 kHz to 1 Hz, for electrodes areas shown in Figure 3.8d. These plots are used to analyse the charge transfer resistance (R_2) and ion transport kinetics. The fitted circuit model and the zoomed image is shown in the inset of Figure 3.8d. The solution resistance (R_1) was found to be $0.7\ \Omega$, indicating minimal resistance from the electrolyte solution to current flow. The charge transfer resistance (R_2), represents the resistance at the electrode/electrolyte interface and governs charge transfer kinetics, and the value was found as $2.6\ \Omega$. This low value of charge transfer resistance suggests rapid charge transfer at the interface. The presence of the constant phase element (Q) indicates non-ideal capacitive behaviour arising from the frequency-dependent response of the electrochemical system. It accounts for factors such as surface roughness, porosity of the active material, and heterogeneity in the electrode coating. Additionally, the Warburg element (W) represents the ion diffusion kinetics within the electrode–electrolyte interface, with its impedance varying proportionally to $\sqrt{\omega}$, where ω is the frequency of the applied AC voltage.

Figure 3.8(e) demonstrates the excellent cycling performance of the coin-cell supercapacitor, exhibiting approximately 93% capacitance retention after 10,000 charge–discharge cycles at $5\ \text{A g}^{-1}$. Furthermore, as illustrated in Figure 3.8(f), the coin-cell supercapacitor delivered an energy density of $11.53\ \text{Wh kg}^{-1}$ at a power density of $0.89\ \text{kW kg}^{-1}$ at a discharge current density of $2\ \text{A g}^{-1}$. This corresponds to an equivalent energy density of $46.12\ \text{Wh kg}^{-1}$ for a single EPD MoS_2 nanosheet electrode, highlighting the promising practical applicability of the material for high-performance energy storage devices.

Table 3.1 presents a comparison of the electrochemical performance of recently reported MoS_2 -based supercapacitor electrodes with my present work. Although several composite and hybrid MoS_2 materials exhibit higher specific capacitance, the MWCE-assisted exfoliated MoS_2 nanosheets fabricated in this study demonstrate competitive electrochemical performance, excellent cycling stability (93% capacitance retention after 10,000 cycles), and successful operation in a symmetric coin-cell configuration. These results indicate that the proposed fabrication approach offers a simple and effective route for developing stable MoS_2 -based supercapacitor electrodes.

Table 3.1 Comparison of the electrochemical performances of recently reported MoS₂-based supercapacitor electrodes

Year	Materials	Method	Performance	Journal
2025	Few-layer MoS ₂ /rGO heterostructure	One-pot hydrothermal method + reduction	346.1 F g ⁻¹ at 1 A g ⁻¹ 99.1% after 10,000 C/D cycles	Zhang, et al. <i>Advanced Composites and Hybrid Materials</i> , 2025,108, 8, 1, 2522-0136
2025	Nanoflowers MoS ₂ /MnO ₂ /CNT	Hydrothermal method	457 F g ⁻¹ at 0.15 A g ⁻¹ 96% capacity after 2000 C/D cycles. Energy density of 37 Wh kg ⁻¹	Chowdhury Anik, et al. <i>Mater. Adv.</i> , 2025,6, 9723
2025	Co-doped metal dichalcogenides (MoS ₂ :WS ₂) on carbon felt electrode	Hydrothermal method	1097.9 F g ⁻¹ at 1.0 mV s ⁻¹ 88.9% capacity after 5000 C/D cycles Energy density, 62.3 Wh kg ⁻¹	Özlem Yağci, et al. <i>Materials Chemistry and Physics</i> , 332, 2025, 130223, 0254-0584
2025	WSe ₂ /MoS ₂ hybrid electrode	Hydrothermal method	1169.53 F g ⁻¹ at 1 A g ⁻¹ 82.80 % after 5000 C/D cycles.	Nafis Ahmad, et al. <i>Inorganic Chemistry Communications</i> , 172, 2025, 113789, 1387-7003
2024	SnO ₂ -MoS ₂ composite	Hydrothermal method + Annelation	415 F ⁻¹ g at 1A g ⁻¹ 92.2 % for 5,000 C/D cycles.	Kinza Rafique, et al. <i>Surfaces and Interfaces</i> , 44, 2024, 103650, 2468-0230
2026	MWCE MoS ₂ nanosheet	MWCE + EPD	324 F g ⁻¹ at 2 A g ⁻¹ 93% after 10,000 C/D cycles at 5 A g ⁻¹ (symmetric coin cell) Energy density 13.9 Wh k g ⁻¹	This work

3.2 Electrodeposition of TMDs

3.2.1 Electrodeposition process of MoS₂

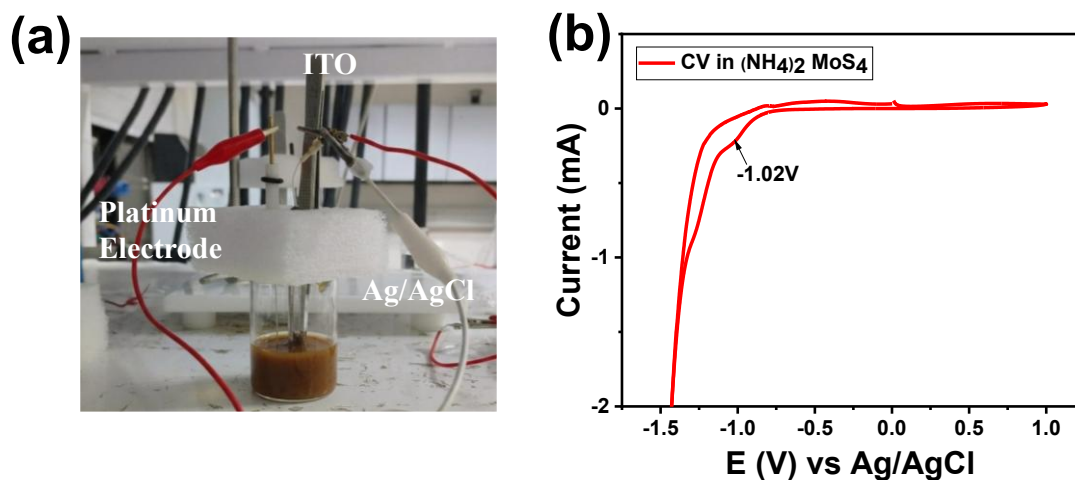


Figure 3.9 (a) Photograph of the three-electrode setup used for the electrodeposition of MoS₂ thin films and (b) CV showing the reduction potential associated with MoS₂ formation at the electrode–electrolyte interface.

The electrodeposition of MoS₂ thin films was carried out as described in Section 2.3. Figure 3.9a shows the three-electrode setup used to determine the reduction potential at which MoS₂ deposition occurs. The CV shown in Figure 3.9b exhibits a reduction peak at approximately -1.02 V versus Ag/AgCl, corresponding to the reduction of the precursor and subsequent formation of MoS₂ on the substrate. Based on this, MoS₂ thin films were deposited by chronoamperometry at an applied potential of -1.02 V for 10 min.

The electrodeposited MoS₂ films were subsequently characterized using SEM and Raman spectroscopy. Figures 3.10a and 3.10b show the SEM images of the MoS₂ thin film deposited on the ITO substrate, revealing a well-covered film morphology. The annealed MoS₂ film exhibits the characteristic in-plane E_{12g} mode at 380 cm^{-1} and out-of-plane A_{1g} mode at 405.7 cm^{-1} corresponding to the 2H phase.

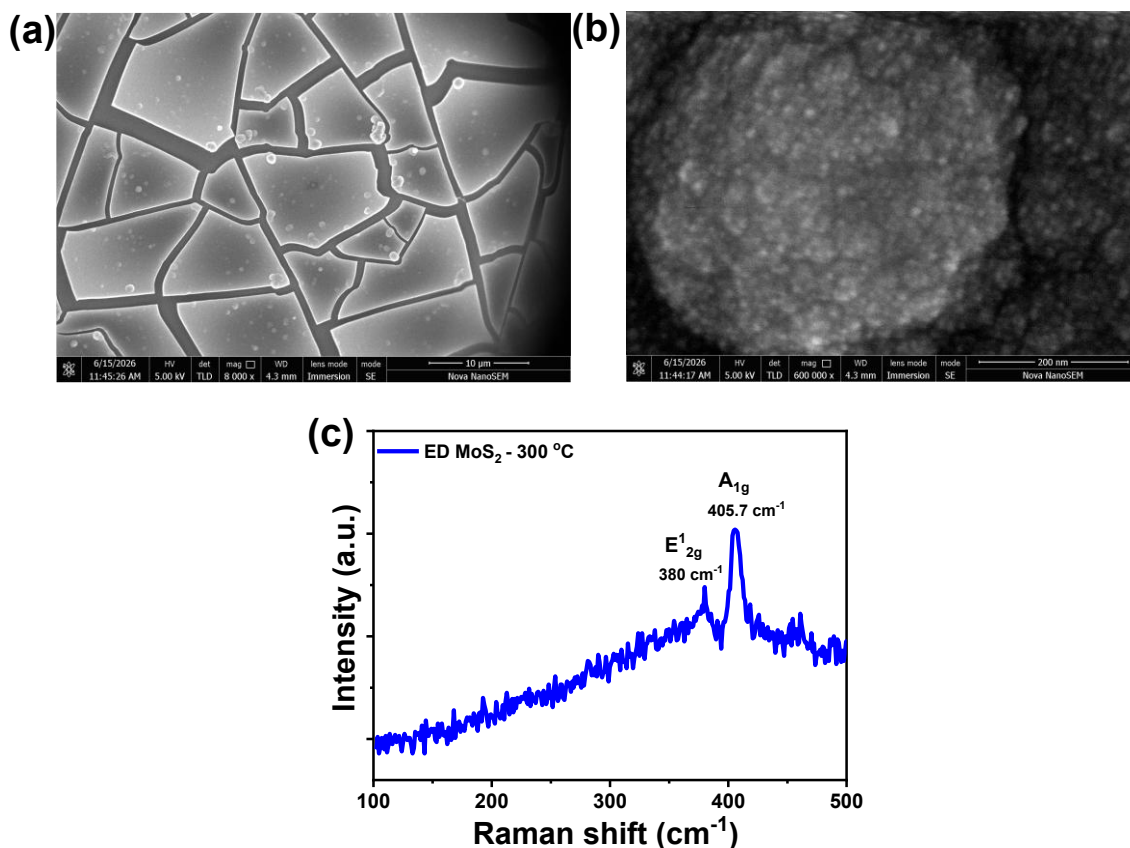


Figure 3.10 (a, b) shows the SEM images of the electrodeposited MoS₂ (c) Raman spectra of the annealed MoS₂ thin film at 300 °C

3.2.2 Electrodeposition process of CoS

The electrodeposition of CoS thin films was carried out similarly, as described in Section 2.4. Initially, CV measurements were performed to determine the suitable deposition potential, followed by electrodeposition at an applied potential of -1.1 V for 10 min. Figures 3.11a and 3.11b show the three-electrode setup used for deposition and the corresponding CV curve, respectively. The electrodeposited CoS thin films were subsequently characterized using Raman spectroscopy and SEM to investigate their structural and morphological properties.

Figure 3.11c shows the Raman spectrum of the electrodeposited CoS thin film, where characteristic peaks are observed at 475, 517, and 676 cm⁻¹, corresponding to the E_g, F_{2g}, and A_{1g} modes of CoS, respectively. Figures 3.10d and 3.10e present the SEM images of the CoS thin film, revealing a highly porous and interconnected network of ultrathin

nanosheet-like structures, which are expected to provide a large electrochemically active surface area.

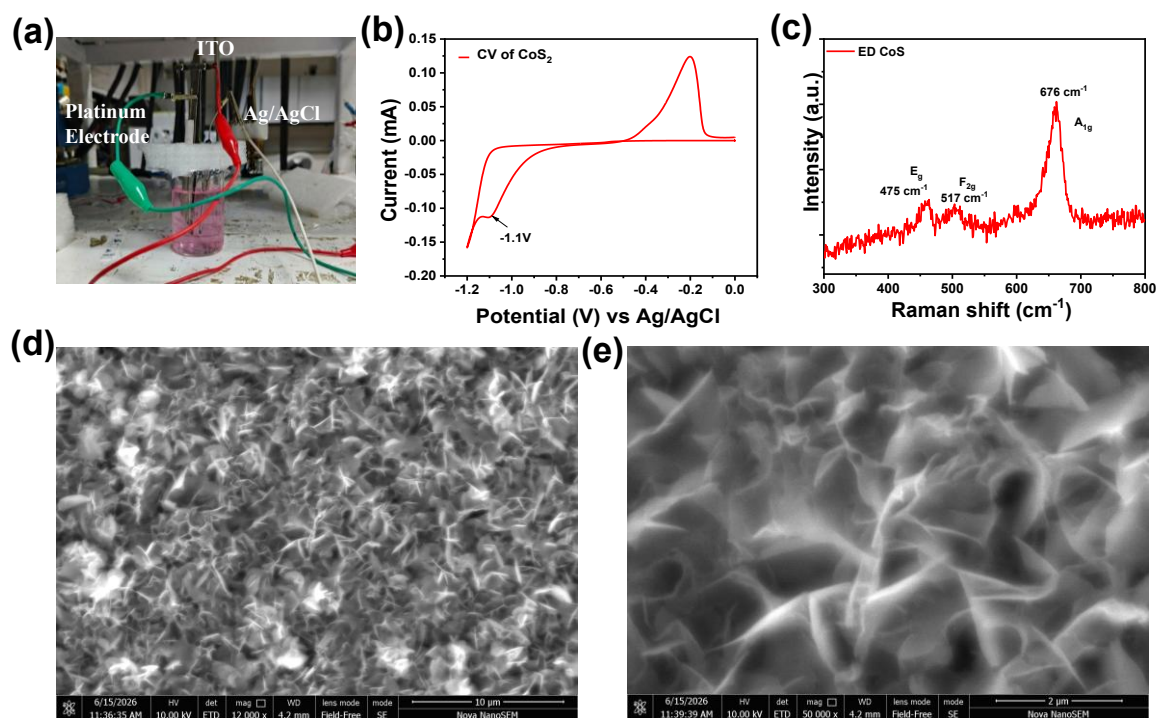


Figure 3.11 (a) Photograph of the three-electrode setup used for the electrodeposition of CoS and (b) corresponding CV curve showing the reduction potential. (c) Raman spectra of CoS film and (d, e) show the SEM images of CoS deposited on ITO.

CHAPTER 4

CONCLUSION AND FUTURE PROSPECTS

4.1 Conclusion

In this work, mixed-phase 1T/2H MoS₂ nanosheets were successfully synthesized through a rapid and scalable microwave-assisted chemical exfoliation method. The exfoliated nanosheets were subsequently deposited onto carbon paper substrates by electrophoretic deposition to fabricate binder-free supercapacitor electrodes with enhanced electrochemical performance. The EPD-MoS₂ electrode exhibited a high specific capacitance of 324 F g⁻¹ at a current density of 2 A g⁻¹ in a three-electrode configuration, demonstrating the effectiveness of the mixed-phase nanosheet architecture for energy storage applications. Furthermore, a symmetric coin-cell supercapacitor based on EPD-grown MoS₂ electrodes was successfully fabricated and demonstrated excellent cycling stability, retaining approximately 93% of its initial capacitance after 10,000 charge–discharge cycles at a current density of 5 A g⁻¹. In parallel, the electrochemical deposition of other TMD materials was optimized, highlighting the versatility of the developed synthesis and deposition approaches for the scalable fabrication of advanced electrode materials. Overall, this study demonstrates the potential of combining microwave-assisted exfoliation with low-cost deposition techniques for the development of high-performance TMD-based energy storage devices.

4.2 Future Prospects

The simple, scalable, and cost-effective synthesis and deposition approaches developed in this work provide several opportunities for future research in advanced energy storage technologies. Future studies can focus on the exploration of TMD-based electrodes for micro-supercapacitor applications. The optimization of material deposition onto interdigitated electrode architectures could enhance ion transport pathways and improve the electrochemical performance of symmetric micro-supercapacitors. In addition, the development of heterostructure TMD electrodes, such as MoS₂/CoS and other hybrid systems, through electrodeposition offers a promising strategy for achieving superior energy storage performance. Such efforts could contribute to the realization of next-generation high-energy and high-power supercapacitor devices.

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